

$$\dots \quad \hat{y}(k|\hat{\underline{\theta}}) = \underline{\psi}^T(k) \hat{\underline{\theta}}$$

where

$$\underline{\psi}^T(k) = [-y(k-1), \dots, -y(k-m), u(k), \dots, u(k-m)]$$

$$\hat{\underline{\theta}}(k) = [\hat{a}_1, \dots, \hat{a}_m, \hat{b}_0, \dots, \hat{b}_m]^T$$

$$\text{goal: find } \hat{\underline{\theta}}^* = \underset{\hat{\underline{\theta}}}{\text{argmin}} \underbrace{\frac{1}{N} \sum_{k=1}^N [\underbrace{y(k) - \hat{y}(k|\hat{\underline{\theta}})}_{e(k|\hat{\underline{\theta}})}]^2}_{J_N(\hat{\underline{\theta}})} = \underset{\hat{\underline{\theta}}}{\text{argmin}} \frac{1}{N} \sum_{k=1}^N e(k|\hat{\underline{\theta}})^2$$

$$\underline{e}_N(\hat{\underline{\theta}}) \triangleq \begin{bmatrix} e(1|\hat{\underline{\theta}}) \\ \vdots \\ e(N|\hat{\underline{\theta}}) \end{bmatrix}$$

$$\text{similarly: } \underline{y}_N \triangleq \begin{bmatrix} y(1) \\ \vdots \\ y(N) \end{bmatrix}$$

Consider

$$\sum_{k=1}^N e(k|\hat{\underline{\theta}})^2 = e(1|\hat{\underline{\theta}})^2 + e(2|\hat{\underline{\theta}})^2 + \dots + e(N|\hat{\underline{\theta}})^2 = \underline{e}_N^T(\hat{\underline{\theta}}) \underline{e}_N(\hat{\underline{\theta}})$$

$$e(k|\hat{\underline{\theta}}) = y(k) - \hat{y}(k|\hat{\underline{\theta}}) = y(k) - \underline{\varphi}^T(k) \hat{\underline{\theta}} \quad (176)$$

$$\underline{e}_N(\hat{\underline{\theta}}) = \begin{bmatrix} e(1|\hat{\underline{\theta}}) \\ \vdots \\ e(N|\hat{\underline{\theta}}) \end{bmatrix} = \begin{bmatrix} y(1) - \underline{\varphi}^T(1) \hat{\underline{\theta}} \\ \vdots \\ y(N) - \underline{\varphi}^T(N) \hat{\underline{\theta}} \end{bmatrix} = \underbrace{\begin{bmatrix} y(1) \\ \vdots \\ y(N) \end{bmatrix}}_{\underline{y}_N} - \underbrace{\begin{bmatrix} \underline{\varphi}^T(1) \\ \vdots \\ \underline{\varphi}^T(N) \end{bmatrix}}_{\underline{\Phi}_N^T} \hat{\underline{\theta}}$$

where $\underline{\Phi}_N = [\underline{\varphi}(1) | \dots | \underline{\varphi}(N)]$.

recall: $\underline{\varphi}(k) = \begin{bmatrix} -y(k-1) \\ \vdots \\ -y(k-m) \\ u(k) \\ \vdots \\ u(k-m) \end{bmatrix} \in \mathbb{R}^{(m+m+1) \times 1}$

$p \times 1$

$$p \triangleq \dim \underline{\varphi}$$

in ARX

$$p = m + m + 1$$

$\Rightarrow \underline{\Phi}_N \in \mathbb{R}^{p \times N}$ matrix with p -rows
 N -columns

$$\Rightarrow \underline{e}_N(\hat{\underline{\theta}}) = \underline{y}_N - \underline{\Phi}_N^T \hat{\underline{\theta}}$$

$$\dots J_N(\hat{\underline{\theta}}) = \frac{1}{N} \sum_{k=1}^N e^2(k|\hat{\underline{\theta}}) = \frac{1}{N} \underbrace{\left[\underline{y}_N - \Phi_N^T \hat{\underline{\theta}} \right]^T}_{= \frac{1}{N} \underline{e}_N^T(\hat{\underline{\theta}})} \underbrace{\left[\underline{y}_N - \Phi_N^T \hat{\underline{\theta}} \right]}_{= \underline{e}_N(\hat{\underline{\theta}})} = \quad (177)$$

$$= \frac{1}{N} \left[\underline{y}_N^T \underline{y}_N - \underbrace{\hat{\underline{\theta}}^T \Phi_N^T \underline{y}_N}_{= (\Phi_N^T \hat{\underline{\theta}})^T} - \underbrace{\underline{y}_N^T \Phi_N^T \hat{\underline{\theta}}}_{= \hat{\underline{\theta}}^T \Phi_N \Phi_N^T \hat{\underline{\theta}}} + \hat{\underline{\theta}}^T \Phi_N \Phi_N^T \hat{\underline{\theta}} \right] = (\dots)$$

Note that

$$\hat{\underline{\theta}}^T \Phi_N^T \underline{y}_N = \underbrace{(1 \times p)(p \times N)(N \times 1)}_{1 \times 1} = \underbrace{(1 \times N)(N \times p)(p \times 1)}_{1 \times 1}$$

these quantities are scalar!
 $\Rightarrow$ "transposing" is useless.

$$\hat{\underline{\theta}}^T \Phi_N^T \underline{y}_N = \underline{y}_N^T \Phi_N \hat{\underline{\theta}}$$

$$J_N(\hat{\underline{\theta}}) = (\dots) = \frac{2}{N} \left[\frac{1}{2} \hat{\underline{\theta}}^T \Phi_N \Phi_N^T \hat{\underline{\theta}} - \underline{y}_N^T \Phi_N^T \hat{\underline{\theta}} \right] + \frac{1}{N} \underline{y}_N^T \underline{y}_N$$

does not affect the minimization. $\Leftarrow$ depends only on $y(1) \dots y(N) \Rightarrow$ "constant" (w.r.t. dataset)

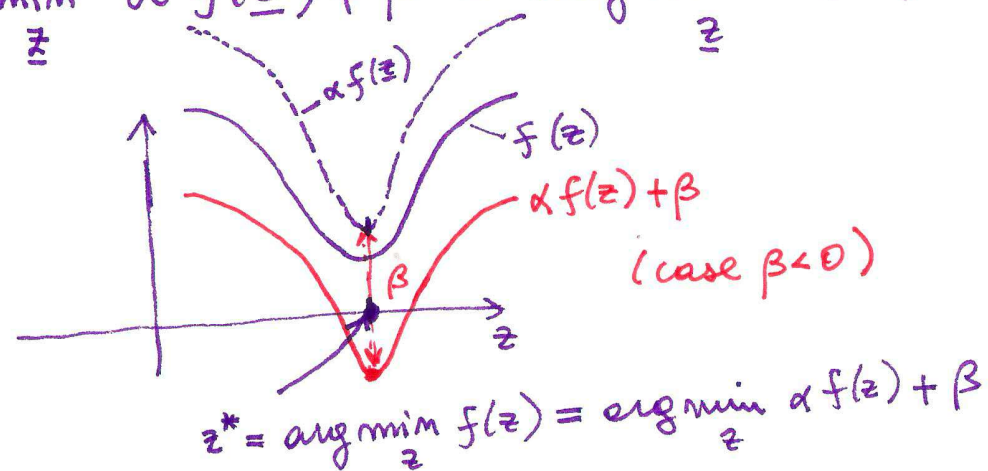
Note that, in general:

(I78)

$$\forall \beta \in \mathbb{R}$$

$$\forall \alpha > 0$$

$$\arg \min_{\underline{z}} \alpha f(\underline{z}) + \beta = \arg \min_{\underline{z}} f(\underline{z})$$



in our case

$$\alpha = \frac{2}{N} > 0$$

$$\beta = \frac{1}{N} \underline{y}_N^T \underline{y}_N$$

$$\underline{z} \rightarrow \hat{\underline{\theta}}$$

$$\Rightarrow \hat{\underline{\theta}}^* = \arg \min_{\hat{\underline{\theta}}} J_N(\hat{\underline{\theta}}) = \arg \min_{\hat{\underline{\theta}}} \left\{ \frac{1}{2} \hat{\underline{\theta}}^T \Phi_N \Phi_N^T \hat{\underline{\theta}} - \underline{y}_N^T \Phi_N^T \hat{\underline{\theta}} \right\}$$

Quadratic form.

Minimization of quadratic forms.

$$f(\underline{z}) = \frac{1}{2} \underline{z}^T \underline{M} \underline{z} - \underline{c}^T \underline{z}$$

$$\underline{M} > 0 \quad \left\{ \begin{array}{l} \text{i.e.} \\ \forall \underline{z} \neq 0 \\ \underline{z}^T \underline{M} \underline{z} > 0 \end{array} \right.$$

(179)

consider

$$\frac{\partial f(\underline{z})}{\partial \underline{z}} = \underline{z}^T \left(\frac{M + M^T}{2} \right) - \underline{c}^T = \underline{z}^T M - \underline{c}^T$$

if $M = M^T$
(symmetric matrix)

dim $\underline{z} = p$;

$$\frac{\partial f(\underline{z})}{\partial \underline{z}} = \left[\frac{\partial f(\underline{z})}{\partial z_1} \dots \frac{\partial f(\underline{z})}{\partial z_p} \right]$$

$$\underline{z}^T M - \underline{c}^T = 0 \iff \underline{z}^T M = \underline{c}^T$$

$$\Rightarrow \underline{z}^* = \arg \min_{\underline{z}} f(\underline{z}) = M^{-1} \underline{c}$$

if $\exists M^{-1}$

in fact:

$$\underline{z}^T M = \underline{c}^T \iff M^T \underline{z} = \underline{c} \iff M \underline{z} = \underline{c} \iff \underline{z} = M^{-1} \underline{c}$$

$\parallel$
M (by Hp) $\uparrow$
if M is invertible.

Note that if $M > 0 \Rightarrow \exists M^{-1}$

$$\Rightarrow \underline{z}^* = \arg \min_{\underline{z}} f(\underline{z}) = M^{-1} \underline{c}$$

in our case

$$\hat{\underline{\theta}} \leftarrow \underline{z}$$

$$\Phi_N \Phi_N^T \leftarrow M$$

$$\underline{y}_N^T \Phi_N^T \leftarrow \underline{c}^T$$

In our case we have $\Phi_N \Phi_N^T = (\Phi_N \Phi_N^T)^T = \Phi_N^T \Phi_N$ (i.e. $M = M^T$) (I 80)

Assume that $\Phi_N \Phi_N^T > 0$

[in general $\Phi_N \Phi_N^T \geq 0$ semi-positive definite]

in fact $\underbrace{\hat{\theta}^T}_{\theta^T} \underbrace{\Phi_N \Phi_N^T}_{\Sigma} \theta \geq 0$ " $= 0$ " only if $\Phi_N^T \theta = 0$

$$\hat{\theta}^* = (\Phi_N \Phi_N^T)^{-1} \Phi_N^T y_N$$

one-shot solution
for identification of
ARX model.

NOTES:

• Note that Φ_N includes values in the data set! (y & u)
 y_N " " " " " "

- One-shot solution available thanks to
 - quadratic cost (least squares)
 - model is linear in the parameters

$$\Phi_N = [\varphi(1) | \varphi(2) | \dots | \varphi(N)] = \begin{bmatrix} -y(0) & -y(1) & \dots & -y(N-1) \\ -y(-1) & -y(0) & \dots & -y(N-2) \\ \vdots & \vdots & \ddots & \vdots \\ -y(-m+1) & -y(-m+2) & \dots & -y(N-m) \\ u(1) & u(2) & \dots & u(N) \\ u(0) & u(1) & \dots & u(N-1) \\ \vdots & \vdots & \ddots & \vdots \\ u(-m+1) & u(-m+2) & \dots & u(N-m) \end{bmatrix} \quad (18)$$

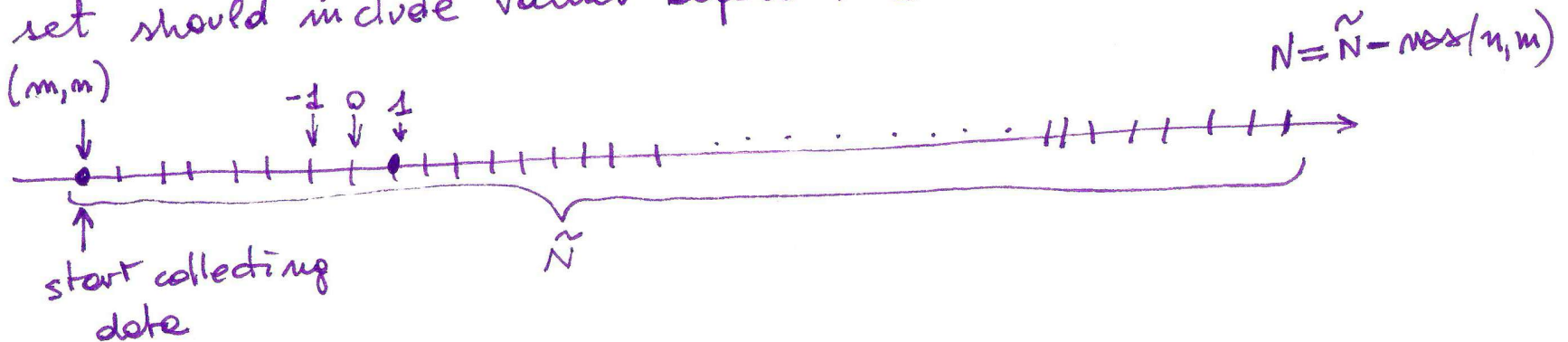
In order to use $\varphi(k)$ for $k=1, 2, \dots, N$

we need ~~$y(0)$~~ $y(-m+1), y(-m+2), \dots, y(-1), y(0), y(1), \dots, y(N)$
before instant 1

and also $u(-m+1), u(-m+2), \dots, u(-1), u(0), u(1), \dots, u(N)$
before inst 1

$\Rightarrow$ the data set should include values before $k=1$

$$1 = \max(m, m)$$

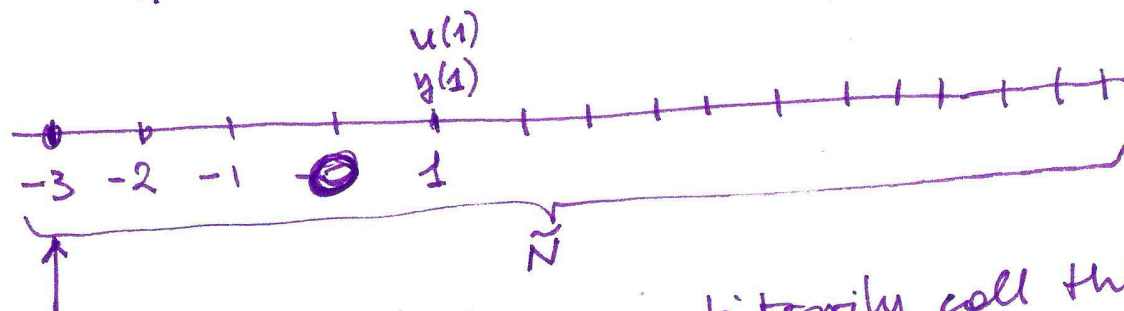


Ex.

$m = 4$

$m = 3$

$1 - \max(m, n) = 1 - 4 = -3$



first data available $\Rightarrow$ arbitrarily call this instant "-3"

If we have measurements for $\tilde{N}$ instants

$$\Rightarrow N = \tilde{N} - \max(m, n)$$

To be clear Data set must include all the inputs & output
to be able to construct $\varphi(1) \dots \varphi(N)$ plus $y_N = \begin{bmatrix} y(1) \\ \vdots \\ y(N) \end{bmatrix}$

we need $y(N)$ too.